

Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

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Motivation: Ownership Changes Beyond Market Structure

- Ownership transitions are common in healthcare.
- IO research often emphasizes **prices, market power, and concentration**.
- But many transitions do **not** change the set of providers in a market.
- In those cases, effects operate through **internal production decisions**: management practices and especially **staffing inputs**.

Research Question

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?
 - ▶ Do responses differ across staff type or organizational structure?
 - ▶ Do responses differ pre vs post-pandemic?
- Why we care: staffing is the primary input into care, so ownership changes can affect **quality** even when market structure is unchanged.

Why Nursing Homes?

- Nursing home care is labor-intensive and heavily regulated.
- **Nursing labor is the main input** into care and a major operating cost.
- Staffing is a natural adjustment margin after ownership change, but it's constrained by:
 - ▶ compliance,
 - ▶ resident outcomes,
 - ▶ demand/occupancy

Nursing Staff Types

- We study four staffing measures:
 - ▶ **RN (Registered Nurses):** clinical assessment, care planning, meds, supervision
 - ▶ **LPN (Licensed Practical Nurses):** routine clinical tasks, meds under RN oversight
 - ▶ **CNA (Certified Nursing Assistants):** direct daily care
 - ▶ **Total nursing:** RN + LPN + CNA
- Ownership change can affect **levels** (total hours) and **mix** (skill composition).

The Cost–Quality Tradeoff in Staffing

- Owners can raise quality through:
 - ▶ **Quantity** of staffing and/or
 - ▶ **Quality / skill mix**
- Either margin increases costs
- But **revenues are relatively flat in the short run**
- So owners face a tradeoff: **higher quality vs. tighter margins.**

Why Staffing Could Rise or Fall After a Transition

- **Cost pressures:** tighter budgets / debt service can push cost cutting.
- **Reorganization:** standardized policies may shift RN/LPN/CNA composition.
- **Quality objective:** owners with stronger quality/compliance goals may increase staffing.
- Net effect is ambiguous $\Rightarrow$ **empirical question.**

Data Sources and Unit of Observation

- **Healthcare Cost Report Information System (HCRIS)**
 - ▶ Annual cost report information
 - ▶ Contains information about costs, revenue, financial data and dates of ownership change
- **Nursing Home Compare (NHC) Archive**
 - ▶ Contains ownership information as well as information on chain affiliation
- **Payroll Based Journal (PBJ)**
 - ▶ Payroll-reported staffing hours by staff type
 - ▶ Used to construct outcome variables
- **Facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024**

Identifying Ownership Changes: Step 1

- **Starting point (HCRIS):** includes a variable indicating **whether and when** an ownership change occurred.
- **Challenge:** ownership is often **layered** (organizations owning organizations), so an administrative flag may not map cleanly to an **economic change in control**.
- **Check:** NHC archive provides owner identities and structure:
 - ▶ **Owner name, ownership level, and owner type**

Ownership Level Challenge

- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
<i>Indirect ownership</i>			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes: Step 2

- **Validate & date using NHC:**

- ▶ Track ownership shares over time within the highest reported ownership level
- ▶ Define a change when **majority control (50%)** transfers between $t - 1$ and t
- ▶ We do not flag changes driven by name changes or within-family reclassification

- **Example from the data:**

- ▶ *Before:*

- ★ ESTES JAMES (84%)
- ★ BURCHFIELD JOHN (10%)
- ★ LEE CLAUDE (6%)

- ▶ *After:*

- ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
- ★ JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)

- ▶ We do **not** treat this as a meaningful ownership change

Dependent Variables: Nurse Staffing (PBJ)

- We study four staffing measures:
 - ▶ **Registered Nurses (RN)**: clinical assessment, medication administration, supervision
 - ▶ **Licensed Practical Nurses (LPN)**: clinical support, medication-related tasks
 - ▶ **Certified Nursing Assistants (CNA)**: direct daily care (bathing, feeding, mobility)
 - ▶ **Total nursing staff**: RN + LPN + CNA
- Outcome is **hours per patient day (HPPD)**: staffing intensity standardized by resident census

$$\text{HPPD}_{s,i,t} = \frac{H_{s,i,t}}{\text{PatientDays}_{i,t}}$$

Final Sample Construction

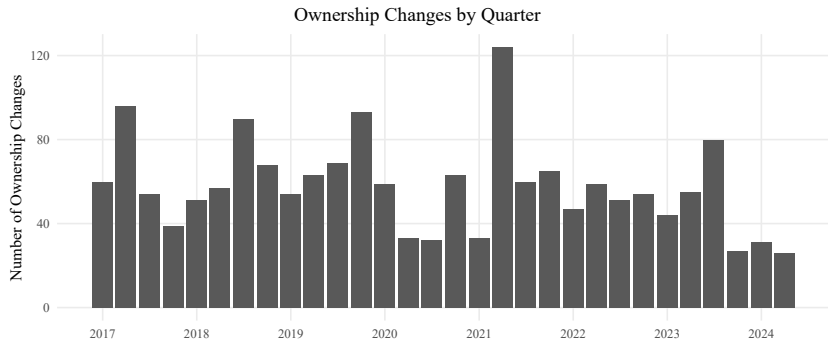
- **Validated ownership-change sample (HCRIS ↔ NHC):**
 - ▶ Keep facilities with **(0,0)** changes in both datasets, or **(1,1)** with dates within 6 months
 - ▶ Event month is the **first month** NHC indicates a change
- **CMS implausible staffing**
 - ▶ RN HPPD = 0 **and** LPN HPPD = 0
 - ▶ Total HPPD < 1.5 or > 12
 - ▶ CNA HPPD > 5.25
- **Additional restrictions:** beds < 15; hospital-based facilities
- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 are treated, about 2% of the final sample

Ownership Change Agreement: NHC vs HCRIS

Cross-tab of Ownership Changes

Count of changes in NHC	2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
	1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
	0	6,217 (46.6%)	499 (3.7%)	15 (0.1%)
		0	1	2+
		Count of changes in HCRIS		

Ownership Changes Over Time (Quarterly)



Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions
- **Variation:** within-facility staffing changes relative to facilities *not yet treated*
- **Event study (dynamic effects):**

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_{\tau} \mathbf{1}\{event_time_{it} = \tau\} \cdot Treat_i + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

- **Key objects:**
 - ▶ Y_{it} : staffing outcome in **HPPD** (RN, LPN, CNA, Total)
 - ▶ β_{τ} : effect at event time τ (relative to the reference period)
 - ▶ α_i facility FE; λ_t month FE; X_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
 - ▶ SEs clustered at the facility level

Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD (headline effect):**

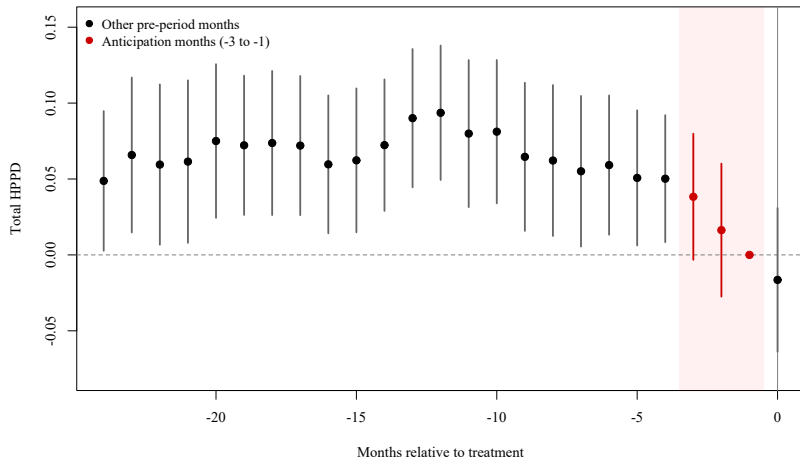
$$Y_{it} = \beta Post_{it} + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- **Key objects:**
 - ▶ $Post_{it} = 1$ in all months after a change (= 0 before)
 - ▶ β : **average post-change effect** on staffing (HPPD)
 - ▶ Same fixed effects and controls as the event-study specification
 - ▶ SEs clustered at the facility level

Identification: Parallel Trends

- **Parallel trends:** absent an ownership change, treated facilities would have followed the same staffing trends as facilities not yet treated.
- **Why anticipation is plausible in ownership transitions**
 - ▶ Changes are not instantaneous: negotiations and operational handoffs precede the recorded month
 - ▶ Staffing can adjust during planning (turnover, hiring freezes, renegotiated contracts)
 - ▶ Recorded dates may reflect administrative/legal timing, not first operational change
- **Implication:** months just before $\tau = 0$ may violate parallel trends.

Evidence of violation in the Pre-Period



Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.
- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period
- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0 : \beta_\tau = 0 \quad \forall \tau < 0$$

- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

Parallel Trends Tests (Joint Wald) — Total Staffing

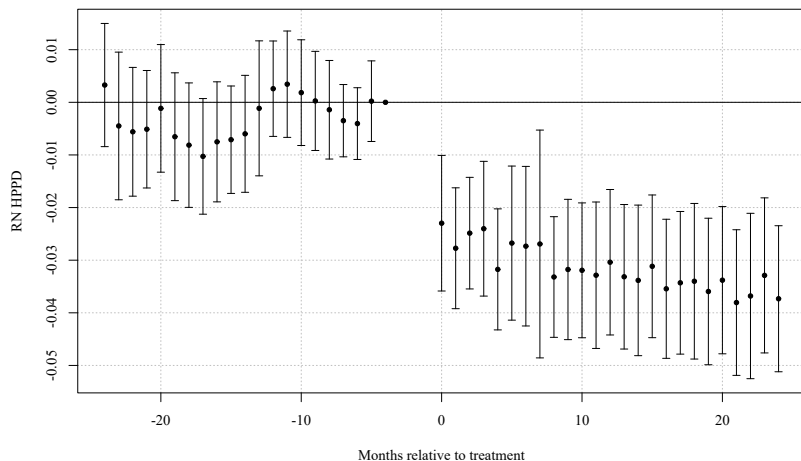
	Total (Levels)	Total (Logs)
2 Year Full Pre-Window	51.22 (23) [0.0006]	49.22 (23) [0.0012]
2 Year Window with Donut	24.92 (20) [0.2046]	22.26 (20) [0.3268]
1 Year Window with Donut	6.08 (8) [0.6387]	6.33 (8) [0.6102]

Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report χ^2 (df) [p-value].

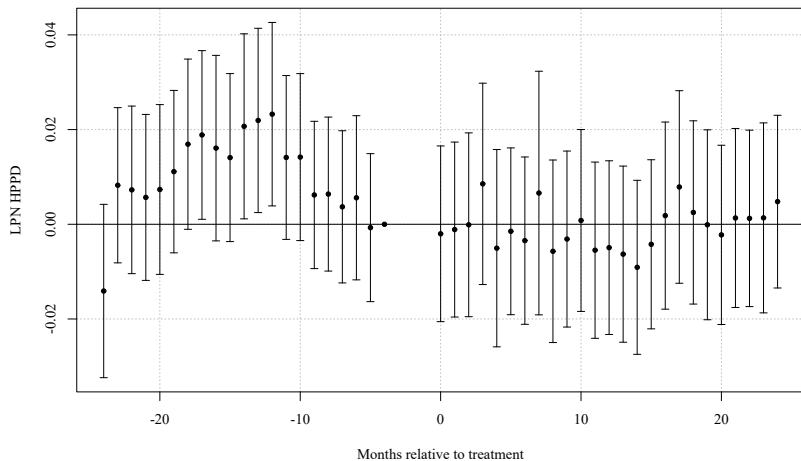
Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

Full outcome-by-staff-type table available in appendix.

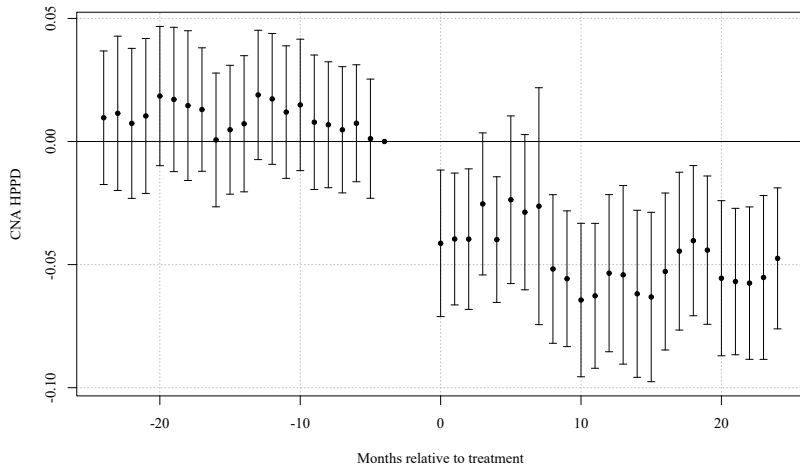
Main Results: Event Study (RN HPPD)



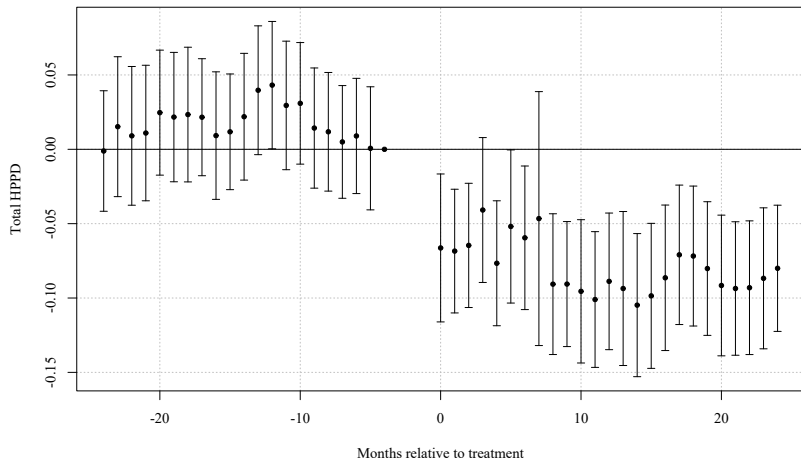
Main Results: Event Study (LPN HPPD)



Main Results: Event Study (CNA HPPD)



Main Results: Event Study (Total Nursing HPPD)



Average Post-Change Staffing (TWFE DiD)

- TWFE DiD summarizes the **average** staffing change after an ownership transition

Table: Two-Way Fixed Effects Estimates of *post* on Staffing Outcomes (Baseline, Without anticipation)

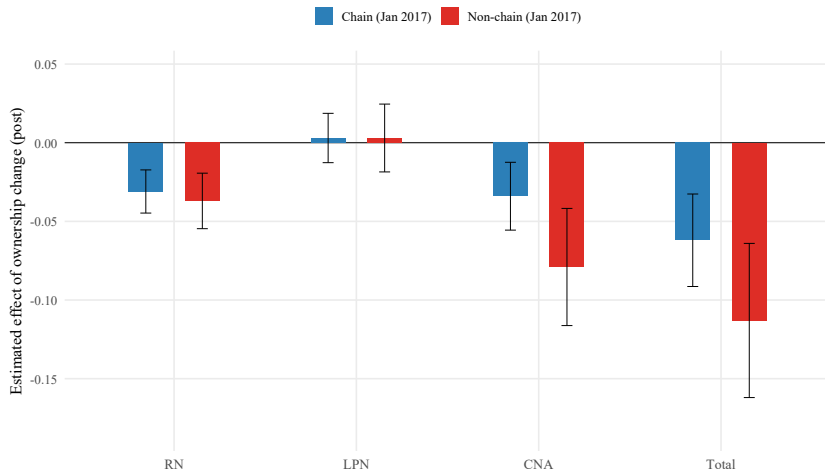
	Outcomes			
	RN	LPN	CNA	Total
HPPD	−0.0320.005***	0.0000.006	−0.0540.009***	−0.0860.013***
Log(HPPD)	−0.0920.016***	0.0030.009	−0.0280.005***	−0.0260.004***

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. The table reports staffing levels (HPPD) and log staffing levels (Log(HPPD)). Sample: Without anticipation ($N_{\text{HPPD}} = 628, 107$; $N_{\text{Log}} = 628, 107$).

All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case mix quartile indicators.

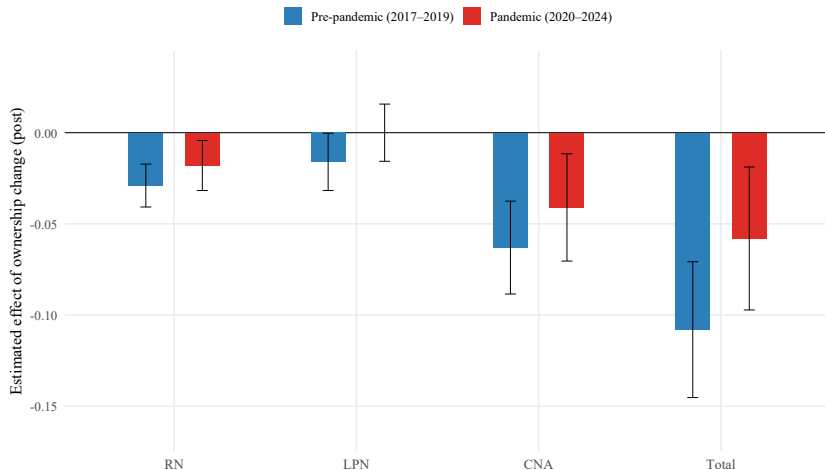
Heterogeneity: Chain vs Non-Chain Facilities

TWFE Post Estimates: Chain vs Non-chain (HPPD, Without Anticipation)



Heterogeneity: Pre vs Pos-Pandemic

TWFE Post Estimates: Pre-pandemic vs Pandemic (HPPD, Without Anticipation)



Robustness: Anticipation and the Donut Window

Additional Robustness Checks (Summary)